

Quiz 10

A typical incoming first-year class at Hood has around 300 students. Imagine the chance experiment of gathering a first-year class. We will think about birthdays of students in the class, making two simplifying assumptions:

- There are 365 possible birthdays; we ignore February 29th.
- Birthdays for different students are independent.

1. What is the probability that at least one student in the 300-member incoming class has my birthday (September 10th)?

The complement of this event is the event that no member of the class is born on September 10th. The probability of that event is $\left(\frac{364}{365}\right)^{300}$ by the assumptions. This probability is thus $1 - \left(\frac{364}{365}\right)^{300} \approx 0.5609077642$. We could also use the Poisson approximation—more important in the days before very precise calculators—which gives $1 - e^{-(300)\left(\frac{1}{365}\right)} \approx 0.5604121995$. Since we have the more precise value, we'll use it.

2. Let N be the random variable giving the number of different birthdays in the 300-member incoming class. What is $E[N]$? Use the method of indicators answer this question. Explain clearly how you are using the method, writing N as a sum of indicator random variables and calculating the expected value of each indicator.

Let A_i be the event that there is at least one member of the class born on the i th day of the year ($i = 1, 2, \dots, 365$). By the same reasoning as in the first question, $P(A_i) = 1 - \left(\frac{364}{365}\right)^{300}$. Let X_i be the indicator variable for the event A_i . As we saw in class, $E[X_i] = P(A_i)$.

We can write N as $N = X_1 + X_2 + \dots + X_{365}$. Therefore, $E[N] = 365 \left(1 - \left(\frac{364}{365}\right)^{300}\right) \approx 204.7$. We expect around 205 different birthdays in the class of 300 students.